

HemaVision-Net: A High-Resolution Convolutional Framework for Severely Occluded Blood Smears

Abstract—The peripheral blood smear (PBS) analysis plays a crucial role in diagnosing hematological disorders, but its manual interpretation is both tedious and suffers from interobserver variations, whereas deep learning techniques have shown great promise, however, despite their excellent performance on benchmarked clean datasets with isolated cells such as the RT DETR model, CNN and transformer-based networks demonstrate poor clinical applicability, where transformers experience high false positives with RBC precision as low as 57.1% based on confusion matrices and low classification accuracy, to solve this problem, we introduce HemaVisionNet with a YOLOv11 architecture that detects and classifies Red Blood Cells, White Blood Cells, and Platelets in crowded PBS samples with a dual-training approach using multiple pretrained data sets and fine-tuning with a high-resolution image size of 800 by 800 pixels, along with optimized data augmentation and higher bounding box loss, we discover the SAHI Paradox that slicing inferences decrease recall to 65.2%, whereas our model obtains an mAP of 0.5 for BCCD at 92.93% and 84.99% on out-of-distribution images, thereby emphasizing its novelty and significance.

Index Terms—Blood cell detection, Red Blood Cells (RBC), White Blood Cells (WBC), Platelets, High-resolution microscopy, Cross-institutional validation.

I. INTRODUCTION

Peripheral blood smear (PBS) analysis is an important method for diagnosing hematological diseases, whereby observing red blood cells (RBCs), white blood cells (WBCs), and platelets, various abnormalities are detected. However, because of these microscopic images often show densely packed cells, overlapping structures, staining variation, and various imaging conditions, accurate analysis of blood smear images is still a major challenge. Furthermore, manual microscopic examination is time-consuming and highly dependent on expert experience, which can lead to interobserver variability and reduce diagnostic consistency [1], [2]. To overcome these limitations, the use of deep learning-based automated blood cell analysis methods has increased significantly in recent years. Among the existing methods, CNN-based detection framework, two-stage detection-classification pipeline, and transformer-based architecture are mainly used. Several CNN-based detection models have demonstrated good performance on benchmark datasets such as BCCD [3]. On the other hand, although transformer-based approaches are able to capture the global contextual relationship of an image, their effectiveness is often limited in microscopic images with densely distributed and overlapping cells [4]. In addition, most studies rely on limited datasets, which reduces the generalization capability of the model under different real imaging conditions [5]. To solve these issues, this study proposes a high-resolution CNN-based

framework named HemaVision-Net, which is designed for RBC, WBC and platelet detection in complex blood smear images. The proposed framework emphasizes localized feature learning to ensure more accurate detection in dense cellular environments. In addition, a two-stage training strategy is used to improve feature generalization and robustness. In addition, the limitations of transformer-based approaches and slicing-based inference methods in hematological image analysis are also analyzed in this study.

The main contributions of this research are as follows:

- A high-resolution CNN-based framework named HemaVision-Net is proposed to improve blood cell detection in densely populated blood smear images.
- A two-stage training strategy is developed to improve feature generalization and detection performance under different imaging conditions.
- A comparative discussion of CNN and transform models is presented for microscopic blood cell analysis.
- The impact and limitations of slicing-based inference techniques in hematological image analysis are examined.
- Validation is performed on internal and external datasets to build the robustness of the model under different imaging conditions.

II. LITERATURE REVIEW

Recent advancement in deep learning have significantly improved automated blood cell analysis using microscope images. Existing approaches can be broadly categorized into object detection based methods and classification based methods.

Detection based approaches aim to localize and classify multiple cell simultaneously. X.Wang et al. [6] proposed a two stage framework combining an improved YOLOv7 detector with EfficientNetV2 for blood cell classification, achieving a mAP@0.5 of 97.17% on the BCCD dataset. However, the two stage architecture increases computational complexity and limits real time applicability. Similarly P.Rani et al [7] introduced a hybrid framework integrating YOLOv9 with MSR-CNN for image enhancement and detection. Although the approach improves visual quality, It lacks standardized evaluation metrics such as mAP, making comparison with state of the art detectors difficult. Transformer based detection models have also gained attention in recent years. X.Cheng et al [8] proposed BCRT=DETR, a light weight transformer based detection model for blood cell detection tasks. While the model demonstrates competitive performance, it lacks extensive benchmarking across multiple datasets. Recent real time detection transformer such as RT=DETR enable end

to end detection without anchor boxes or non maximum suppression, improving inference speed while maintaining high accuracy. However most transformer based detection models are evaluated on limited datasets restricting their generalizability capability.

Other detection related works include X. Zheng et al [9], who developed MCDAF-Net, a multi scale cross deformation attention fusion network for white blood cell detection, achieving mAP values around 93-95%. Despite strong performance, the model is evaluated on a single dataset and lacks real time performance analysis. In addition, V.M Prathap and S.Yadav [10] proposed a YOLO-based framework for detecting Plasmodium Vivax in blood smear images, demonstrating robustness under challenging condition such as overlapping cells. However, this approach is disease specific and not generalized for multi class blood cell detection. On the other hand, several studies focus primarily on classification rather than detection. H. Zhu et al. [11] introduced BccT, a transformer based model incorporating token fusion to reduce redundancy and improve classification efficiency. Similarly A. Shahzad et al [12] proposed a vision transformer based approach enhanced with PCA-NCA feature selection, achieving strong classification accuracy on benchmark datasets. However these methods do not address object localization and therefore cannot provide detection metrics such as mAP. CNN based classification approaches have also showed high performance. K.T.N et al [13] developed a convolutional neural network based framework achieving classification accuracy above 95% on blood smear datasets. Likewise M.T. Ullah et al proposed a hybrid model combining DenseNet121, mobilenetv2 and resnet101 with a fuzzy decision system achieving accuracy of 99.79%. These approaches are limited to classification tasks and lack evaluation on detection scenarios. In addition to model development, dataset contribution play a crucial role in advancing this field. S.S Bousheri et al [14] introduced a large scale annotated peripheral blood dataset with over 40,000 images, supporting deep learning based hematological analysis. However, the dataset has not been extensively evaluated using modern detection architectures. Zhijie et al [15] introduced a hierarchical multi-scale neural network framework for schistocyte detection in blood smears. The lightweight combination of the cross scale dynamic interaction detection head and normalized Wasserstein distance maintains a mAP@0.5 of 0.946.

From the above literature, several limitations can be identified. Most existing works focus on classification rather than detection, ignoring precise localization of blood cells. Detection models are largely dominated by YOLO based approaches which require anchor tuning and post processing steps. Transformer based detection models are still under explored and lack comprehensive multi dataset evaluation. Many studies rely on single datasets, which limit the generalizability of the model in different imaging conditions. There is a lack of standardized benchmarking using mAP in multiple datasets.

III. METHODOLOGY

The primary objective of this research is achieving robust high precision localization and classification of Red Blood Cell (RBC), white blood cell (WBC) and platelets in highly dense, occluded microscope smears. Since standard global context architecture struggles with extreme cell density, we propose HemaVision-Net: a customized two stage Convolutional Neural Network (CNN) optimized for high resolution domain adaption and penalized bounding box regression. Furthermore, we conduct an architectural ablation study comparing Vision Transformer (RT-DETR) against our CNN, analyzing the failure points of Slicing Aided Hyper Inference (SAHI). Fig. 1 illustrates the complete methodology workflow.

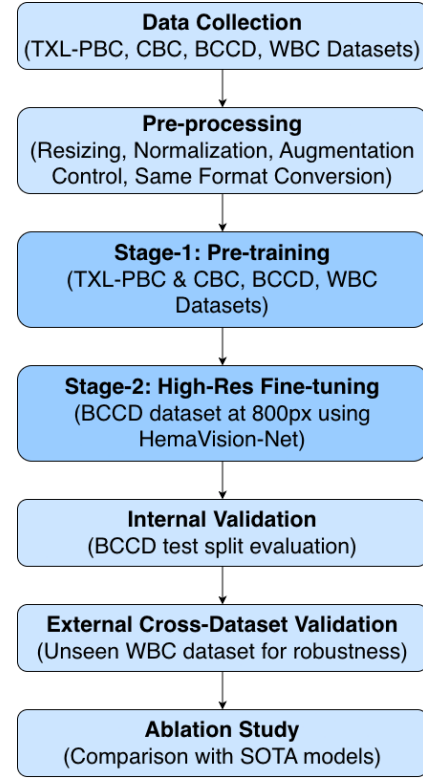


Fig. 1. Complete workflow of the proposed HemaVision-Net methodology highlighting the change from pre-training on different datasets to high resolution specialized fine tuning

A. Dataset Procurement and Arrangement

To guarantee the network acquires infer morphological feature before processing dense groups, we utilized a multi-dataset process which separates data into pre-training, fine-tuning and external validation set.

1) *Pre-training datasets*: We merged two public datasets for feature extraction. TXL-PBC dataset gives high divergence annotations to acquire edge detection and leukocyte structures and CBC dataset provides a standard for counting under various conditions.

2) *Fine tuning and external validation*: For domain adaption, we used the BCCD dataset, featuring severe RBC occlusion mimicking real world smears. A 10% was isolated for internal testing. Finally, the WBC Object Detection Dataset (Kaggle) was kept completely unseen during training which serves as our benchmark for cross-institutional robustness.

B. Proposed Architectural Framework: HemaVision-Net

Vision transformer applied universal self attention, our study with RT-DETR illustrated universal context which becomes a bottleneck in compactly packed smears and it significantly dropped precision. to solve this issue, we designed HemaVision-Net in native YOLOv11 architecture. Fig. 2 presents the internal architecture of the HemaVision-Net framework By utilizing C2f modules and a Path Aggregation Network (PANet) neck, our CNN leverages strictly localized receptive field to distinctly separate touching RBC boundaries without distraction from the global image context.

C. Two Stage High Resolution Training Strategy

Standard detectors train at 640×640 pixels. Compressing a 1920×1080 smear merges adjacent RBCs into distinguishable blobs, forcing the model to approximate boundaries. To solve this, we implemented a two stage strategy. Table I details the customized hyper-parameters utilized for domain adaption.

The model was initially pre-trained on the TXL-PBC and CBC datasets at 640×640 for rapid convergence of morphological features. The Pre-trained weights were then transferred to the BCCD dataset, where the input resolution was increased to $\text{imgsz} = 800$. Crucially, MixUp augmentation was disabled ($\alpha = 0.0$) because, while generally beneficial, it produces biologically implausible overlapping artifacts (“ghost cells”) in hematology.

TABLE I
SURGICAL HYPERPARAMETER CUSTOMIZATIONS FOR DOMAIN ADAPTATION

Parameter	Default YOLOv11	HemaVision-Net	Justification
Input Resolution	640	800	Preserves boundary gradients between overlapping cells
MixUp (α)	0.1	0.0	Avoids biologically invalid overlapping artifacts
Box Loss (λ_{box})	7.5	10.0	Enforces tighter bounding box alignment
Mosaic	1.0	1.0	Supports learning across varying cell densities

D. Loss Function Optimization

To enforce strict boundary adherence, the total loss (L_{total}) combines classification loss (L_{cls}), Distribution Focal Loss (L_{DFL}) and Complete Intersection over Union (L_{CIoU}). Eq. 1 formulates this relationship:

$$L_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (1)$$

By raising the box loss multiplier (λ_{box}) to 10.0, the CIoU penalty forces ranking of the Euclidean distance (ρ) between anticipated and reality centers. This efficiently distinguishes merged RBC clusters, updating placement accuracy. Fig. 3 illustrates the Stage-2 training and validation loss curves which signifies this improving process.

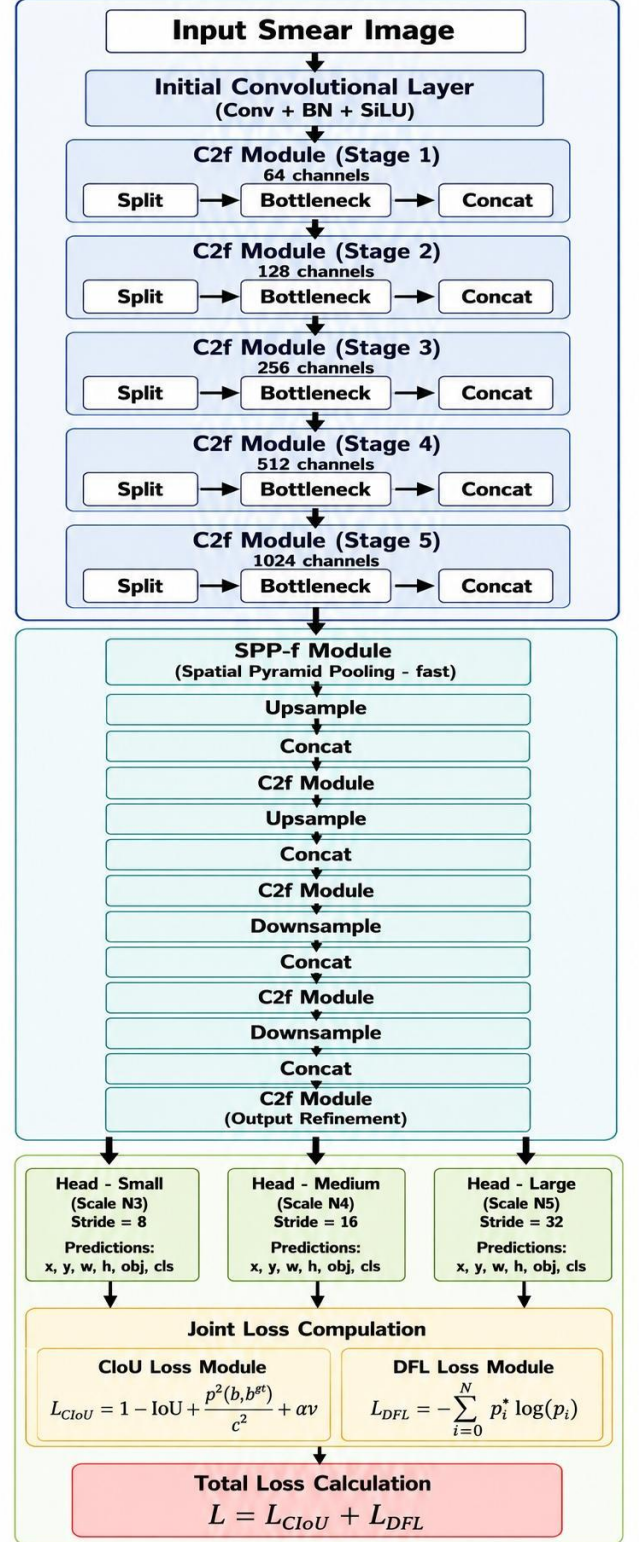


Fig. 2. Architecture of the HemaVision-Net framework, illustrating the localized feature extraction pathways and the customized regression head.

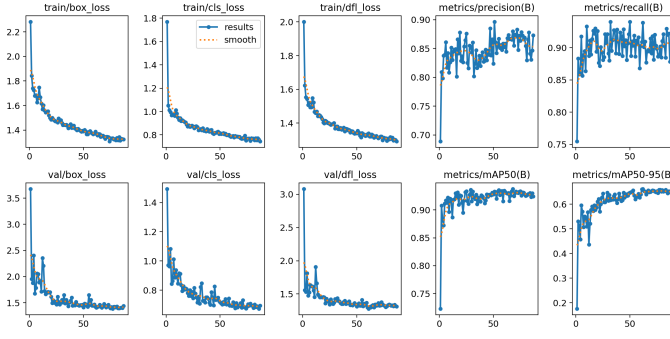


Fig. 3. Training and validation loss curves representing the Stage-2 optimization process.

E. Architectural Limits: Exposing the “SAHI Paradox”

Slicing-Aided Hyper Inference (SAHI) divides high-resolution images into overlapping patches for localized inference. We hypothesized SAHI would improve dense smear detection. However, applying SAHI surprisingly reduced the mAP of RT-DETR from 89.6% to 84.3%, and HemaVision-Net from 92.9% to 84.0%. We term this degradation the “SAHI Paradox”, driven by three limitations:

- **Edge Truncation:** Slicing introduces artificial boundaries cutting through continuous cells. Truncated regions are misinterpreted as independent cells, inflating False Positives.
- **Contextual Scaling Loss:** Patch-wise inference removes global spatial context, disrupting scale consistency.
- **Non-Maximum Merging (NMM) Failure:** Post-processing frequently merges distinct adjacent cells across slice boundaries, dropping Recall to 65.2%.

Consequently, our proposed HemaVision-Net relies entirely on native high-resolution single-pass inference. Fig. 4 demonstrates the accurate, localized bounding box predictions achieved using this framework without SAHI. By bypassing the systemic limitations of grid-based slicing, this unified approach ensures highly accurate, localized predictions and establishes a robust framework for real-world clinical applicability across varying cellular densities.

IV. RESULTS AND DISCUSSION

A. Evaluation metrics

The achievement of the suggested model was analyzed using standard COCO metrics, including Precision (P), Recall (R) and mean average precision (mAP). Eq. 2 calculates the ratio of accurately forecasted positive study to the total forecasted positives which efficiently regulates false-positive detections.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Eq. 3 ensures that the model accurately detects obscured cells within dense clusters.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

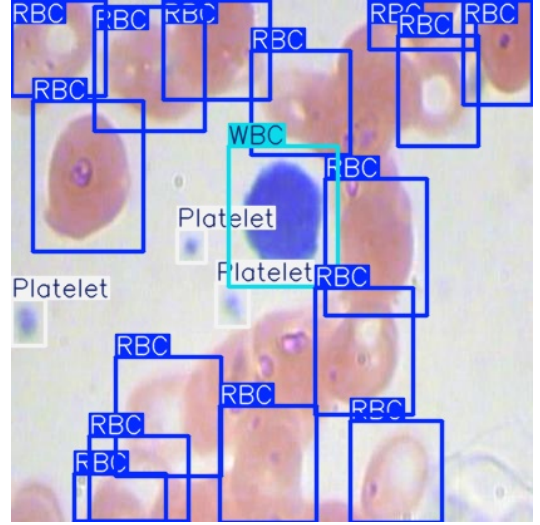


Fig. 4. Localized bounding box predictions demonstrating accurate detection of dense RBC clusters and isolated WBCs using HemaVision-Net without

The initial evaluation metric used in this experiment is mAP@0.5, which calculates the mAP across all classes by averaging the Area Under the Curve (AUC) of the Precision-Recall curve at the Intersection over Union (IoU) threshold of 0.50.

B. Quantitative Evaluation on Internal Test Set

The proposed HemaVision-Net framework was analyzed on the separated 10% internal test split of the BCCD dataset. Right here the main barrier lies in the dense occlusion of (RBCs). Fig. 5 shows the precision-recall curve which proves that our tailored framework accomplished a total mAP@0.5 of 92.93%. Especially the model accomplished close-perfect classification for WBCs resulting an mAP@0.5 of 99.3% and a recall score of 1.0. Platelet detection stabilized at 90.3%. For highly occluded RBCs, the localized receptive fields of the CNN effectively distinguished boundaries, resulting in an RBC mAP@0.5 of 89.2% and a precision of 73.9%. Fig. 6 shows the normalized confusion matrix, further confirms this robust class performance.

C. Cross-institutional Robustness (External Validation)

To guarantee real world clinical applicability, models must be generic across datasets collected under various laboratory conditions which includes variation in staining protocols and microscopy devices. Unlike many existing approaches that validate performance solely on homogeneous splits, we evaluated our final model on a completely independent external dataset (WBC Detection Dataset from Kaggle).

Without prior exposure to its specific staining variations or magnification scales, HemaVision-Net achieved an mAP@0.5 of 84.99%. This minimal degradation across domain shifts confirms that the framework successfully learned fundamental biological morphologies rather than overfitting to dataset-specific artifacts.

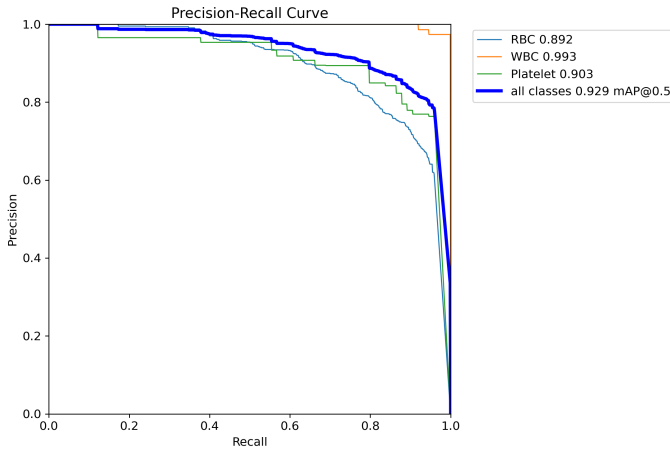


Fig. 5. Precision-Recall (PR) curve illustrating class-wise detection performance of HemaVision-net on the internal test set.

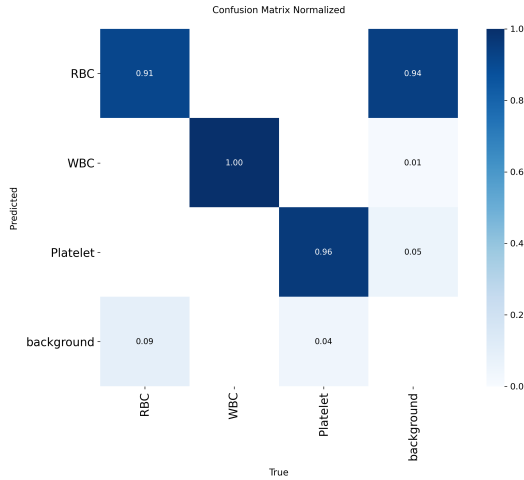


Fig. 6. Normalized confusion matrix showing high classification accuracy, specially for White Blood Cells (WBCs)

D. Ablation Study

To rigorously justify our architectural choices and inference strategies, an ablation study was conducted. Table II summarizes the quantitative comparisons of this analysis.

1) *Architectural Comparison: Transformer vs CNN:* We established a baseline using RT-DETR, a Vision Transformer-based detector. Table II describes that RT-DETR achieved 99.5% accuracy in isolated WBCs, it struggled severely with RBCs occlusion which yielded an overall mAP of 89.6%. After shifting to HemaVision-Net, it is seen that overall mAP mAP@0.5 to 92.93% has improved, showing that CNNs handle highly occluded boundaries better than global-attention mechanisms.

E. Impact of Test-Time Augmentation (TTA) vs Scaling

Table II proves that increasing the inference resolution of RT-DETR to 1024px significantly reduced its efficiency

to 83.18% which indicates spatial context unsteadiness. In contrast, standard inference at 800px combined with Test-Time Augmentation (TTA) for HemaVision-Net preserved the high mAP@0.5 of 92.93% while improving localized recall.

1) *The SAHI Paradox:* SAHI was applied to both architectures to evaluate its impact on dense object detection. Table II shows that SAHI consistently reduced performance, contrary to expectations. The RT-DETR mAP dropped to 84.3% and HemaVision-Net dropped to 84.0% with recall decreasing to 65.2%. This justifies our strategic avoidance of patch-based inference.

TABLE II
ABLATION STUDY ON ARCHITECTURE AND INFERENCE STRATEGIES

Architecture	Strategy	mAP@0.5	RBC	WBC
RT-DETR	Standard (640px)	0.896	0.829	0.995
RT-DETR	High-Res (1024px)	0.831	0.756	0.902
RT-DETR	SAHI (256)	0.843	—	—
HemaVision-Net	Standard (800px)	0.929	0.892	0.993
HemaVision-Net	SAHI (512)	0.840	—	—

F. Discussion

For dense, repetitive structures such as overlapping RBCs, Vision Transformers tend to generate false-positive detections due to global attention over redundant patterns. In contrast, CNN-based HemaVision-Net leverages localized receptive fields to accurately delineate overlapping boundaries. The increased bounding box penalty ($\lambda_{box} = 10.0$) further enforces this precise localization. The ablation study reveals a critical limitation of SAHI. Unlike discrete objects in aerial imagery, blood smears contain continuous, interconnected cells. Grid slicing truncates these biological structures, leading to false detections and the loss of contextual scale. This proves that native high-resolution inference is scientifically superior for hematological applications. HemaVision-Net demonstrates strong cross-institutional robustness, achieving 84.99% mAP on unseen data. The strict avoidance of unrealistic augmentations ensures clinically reliable predictions, making the framework highly suitable for deployment in diverse laboratory environments without site-specific retraining.

G. Limitations

Although this study shows solid performance, several limitations remain. Firstly, high resolution inference (800px) increases computational requirements (VRAM) which can limit real time deployment on low resource clinical edge devices. Secondly, the current model is limited to three primary classes (RBC, WBC, Platelets) and does not classify leukocyte subtypes or pathological variations (e.g., sickle cells). Future work will analyze case segmentation approaches to collect fine grained morphological abnormalities without offering computational efficiency.

V. CONCLUSION

This study presents a high-resolution convolutional framework named HemaVision-Net, which is capable of solving the problems of blood cell detection and classification in densely

located and overlapping peripheral blood smear images. While many existing methods rely on global attention mechanisms, the proposed model emphasizes localized feature learning, which helps to more accurately identify the boundary of overlapping cellular structures. In addition, the combination of two-step training strategy and high-resolution fine-tuning enhances the generalization ability of the model, enabling it to work effectively under various imaging conditions. According to the experimental results, HemaVision-Net achieved 92.93% mAP@0.5 performance in the internal test set and 84.99% performance in the external validation set, which ensures its stability and usability in various institutions. Comparative analysis shows that CNN-based architecture is more effective than transformer-based models in densely populated biological environments, especially in Red Blood Cell detection. In addition, this study introduces the concept of “SAHI Paradox”, which shows that using slicing-based inference can lead to loss of important information due to the boundary breaking of continuous biological structure. This points to an important limitation of some conventional inference methods in medical image analysis. Overall, the proposed HemaVision-Net framework provides a reliable and efficient solution for automated blood cell analysis. It has considerable potential for real clinical use, especially where high accuracy and stable performance are required.

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